# Operational Intelligence Evidence Pack

Benchmark rubric, review model, control comparisons, and falsification criteria for Operational Intelligence.

## Evidence Question

What evidence would convince another experienced engineer that this operating model is useful? Operational Intelligence becomes stronger when claims can be tested, challenged, compared, and revised.

## Review path

Structured practitioner review is captured through /contact so SRE, architecture, AI engineering, governance, executive, and product reviewers can challenge the doctrine with specific evidence requests.

## Evidence Classes

| Class | Purpose | Strong signal |
|---|---|---|
| Implementation example | Show end-to-end behavior | Public-safe inputs, outputs, states, and decisions |
| Benchmark fixture | Make behavior repeatable | Prompt, context, expected behavior, refusal, citation |
| Control comparison | Separate value from baselines | Dashboard-only, chatbot-only, ticket-only, and OI workflow compared |
| Practitioner review | Test real-world comprehension | Structured review by SRE, architect, AI engineer, governance, executive |
| Regression history | Show maturity over time | Failures tracked and corrected |
| Negative case | Test boundaries | Refuses confidential, unsupported, or unsafe requests |

## OI-ROOM-001 Benchmark Rubric

| Dimension | Acceptance question |
|---|---|
| Retrieval | Did the system retrieve doctrine, reference architecture, and relevant OI-ROOM-001 context? |
| Grounding | Are all material claims supported by approved public content? |
| Evidence attribution | Are observation, inference, contradiction, missing evidence, and confirmed fact separated? |
| Hypothesis lifecycle | Do hypotheses move through explicit states? |
| Contradiction handling | Does contradictory evidence change confidence or route? |
| Decision safety | Are actions framed as reviewable options? |
| Operator control | Are consequential actions gated by explicit human approval? |
| Replay reproducibility | Can the scenario be replayed from approved context? |
| Refusal | Does the system refuse confidential or unsupported questions? |

## Evidence Ledger

| Claim | Classification | Result | Next improvement |
|---|---|---|---|
| OI focuses on the decision path from evidence to action. | Original synthesis | Mixed | Collect SRE and observability practitioner feedback |
| The ten-layer model can guide implementation. | Original synthesis | Not independently tested | Review a second public-safe implementation example |
| Evidence graphs improve reviewability. | Derived | Plausible | Compare against chatbot-only and dashboard-only explanations |
| Replay seeds support regression testing. | Derived | Partial | Add replay-backed workflow fixtures |
| Operator control should gate consequential actions. | Established governance principle applied to OI | Strong | Add approval class examples |

## Minimum Conformance Checklist

| Requirement | Observable proof | Failure signal |
|---|---|---|
| Evidence before conclusion | Recommendations cite approved evidence or missing evidence. | Fluent RCA with no source trail. |
| Transaction context | Affected customer, process, or business journey is identified. | Local metric treated as the whole incident. |
| Topology boundary | Dependency, ownership, blast radius, and freshness are visible. | Stale service map treated as fact. |
| Evidence typing | Observation, inference, contradiction, missing evidence, and fact are separated. | Contradictions hidden in prose. |
| Hypothesis lifecycle | Competing hypotheses move through explicit states. | First plausible explanation becomes final answer. |
| Evaluation gate | Grounding, citations, refusal, contradictions, and action safety are tested separately. | Single aggregate score implies trust. |
| Decision packet | Action includes risk, reversibility, owner, fallback, and approval class. | Assistant recommends action without review context. |
| Operator control | Approval, escalation, override, and refusal boundaries are explicit. | Automation boundary is hidden. |
| Replay seed | Scenario can be reproduced from approved context and expected behavior. | Demo cannot be rerun after changes. |
| Learning loop | Reviewed outcome updates memory, patterns, docs, and fixtures. | Learning remains unstructured narrative. |

## Control Comparison Template

| Criterion | Dashboard-only | Chatbot-only | Operational Intelligence |
|---|---|---|---|
| Evidence trail | Signals visible but reasoning is external | Often summarized without stable receipts | Claims cite typed evidence and unknowns |
| Hypotheses | Human-maintained outside the tool | May collapse to first plausible answer | Lifecycle states and falsification criteria are visible |
| Contradictions | Require manual noticing | May be hidden by fluent narrative | Contradictory evidence is first-class |
| Action safety | Handled by process outside dashboard | Risk of over-suggesting action | Decision packet names risk, owner, and approval class |
| Learning | Postmortem and memory are separate | Conversation history is weak memory | Reviewed outcome becomes replay, fixture, and memory |

## Practitioner Review Rubric

| Reviewer | Question they should be able to answer | Weak signal |
|---|---|---|
| SRE | Where does the model improve investigation without replacing incident command? | Cannot distinguish OI from observability or AIOps |
| Architect | Can two teams implement the contracts similarly? | Layer boundaries are interpreted incompatibly |
| AI engineer | Which gates prevent unsafe model behavior? | Only aggregate scores or confidence prose are shown |
| Governance reviewer | Where are approval, audit, refusal, and data boundaries enforced? | Autonomy scope is implicit |
| Executive | What evidence would justify further investment? | Only narrative claims, no benchmark or feedback path |

## Weakening Conditions

The doctrine should be revised if experienced SRE teams cannot distinguish it from existing practice, if independent teams interpret the layers incompatibly, if evidence graphs add complexity without clarity, or if evaluation gates fail to catch obvious regressions.